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Department of Information Science and Engineering

Academic year 2025-2026 (Odd Semester)

Course	ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING		
Date	16th Dec 2025	Maximum Marks	60
Course Code	IS353IA	Duration	120 Min
Sem	V	CIE – II	

Sln	Part A	M	BT	CO
1.1	What is heuristic function in search algorithms?	1	1	2
1.2	State one condition under which Alpha–Beta pruning occurs	1	2	2
1.3	In a 1-NN classifier, if two neighbors are at the same distance but have different labels, how is the final decision made? Give two possible strategies.	2	3	1
1.4	State the fundamental assumptions that Naive Bayes uses regarding the features.	2	2	3
1.5	Justify that random forests outperform bagging when features exhibit strong correlation.	2	3	1
1.6	Write any two difference between MIN and MAX in a minimax game tree	2	1	2

	Part B			
1.a	Explain in detail how A* search works with an example. Show calculations of f(n) = g(n) + h(n) at each step.	5	2	1
1.b	Consider the following tree. Using alpha-beta purning, determine the nodes to be pruned and final value of the root. MAX / \ MIN MIN / | \ / \ 4 8 2 5 1	5	3	2
2.a	Consider the following dataset of **10 customers** where the class label indicates whether the customer will **Buy = Yes/No**.	10	4	3

Tid	Age Group	Marital Status	Annual Spending (K)	Buy
1	Young	Single	55	No
2	Young	Married	70	No
3	Middle	Single	80	Yes
4	Senior	Married	90	Yes
5	Middle	Divorced	65	Yes
6	Senior	Single	60	No
7	Young	Divorced	95	Yes
8	Middle	Married	50	No
9	Senior	Divorced	85	Yes
10	Young	Single	45	No

	Classify the record X=(Age Group = Middle,Marital Status = Divorced,Spending = 75K) as **Buy = Yes** or **No** using the Naive Bayes classifier with Gaussian PDF for Spending.			
3.a	Explain the concept of logistic regression. Discuss its mathematical formulation, working, advantages and limitations.	5	2	3
3.b	Compare bagging, boosting, and random forests algorithms.	5	2	3
4.a	Consider the data set :	10	3	3

x	0.05	0.15	0.25	0.35	0.45	0.55	0.65	0.75	0.85	0.95
y	1	1	1	−1	−1	−1	−1	1	1	1

Below are the Two bootstrap samples are drawn:

1. **Bootstrap 1:** {0.05, 0.05, 0.15, 0.25, 0.35, 0.35, 0.45, 0.75, 0.95, 0.95}
2. **Bootstrap 2:** {0.15, 0.25, 0.25, 0.35, 0.45, 0.55, 0.65, 0.85, 0.85, 0.95}

A decision stump classifier uses the rule: Predict **+1** if $x < 0.30$ Predict **−1** otherwise.

With this information answer the following questions:

argmax $(C_i = 1)$

i. Discuss the concept of bootstrap sampling and decision stump in bagging.
ii. Explain how bagging reduces variance for unstable learners.
iii. If a decision stump splits incorrectly due to noise, explain how bagging helps correct it.
iv. Compare bagging with a single decision stump model.
v. For the test input x = 0.40, compute: Prediction from stump trained on Bootstrap 1 and 2.
vi. Final bagged classifier prediction by majority vote.

5a	Consider the following data set , use a **decision stump** of the form: Predict **+1** if x < k, Predict **−1** if x ≥ k. Perform the following using the Ada Boost algorithm.	10	3	3

$\log\left(\frac{1-\varepsilon_i}{\varepsilon_i}\right)$

i. Initialize weights.
ii. Try stump thresholds k = 3, 4, 5 → choose stump with lowest error.
iii. Compute weighted error
iv. Compute stump weight α.
v. Update sample weights.
vi. Normalize new weights

x	y (true class)
2	+1
3	+1
4	−1
5	−1
6	+1

Course Outcomes: After completing the course, the students will be able to:-	
CO 1	Explain and apply AI and ML algorithms to address various requirements of real-world problems.
CO 2	Design and develop AI and ML solutions to benefit society, science, and industry.
CO 3	Use modern tools to create AI and ML solutions.
CO 4	Demonstrate effective communication through team presentations and reports to analyze the impact of AI and ML solutions on society and nature.
CO5	Conduct performance evaluation, modeling, and validation of AI and ML solutions benefiting lifelong learning.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	8	20	22	10	-	7	29	18	6	-	-